ASINTHA ARIYARATHNE

ELECTRICAL AND ELECTRONIC ENGINEERING GRADUATE



PERSONAL INFORMATION

Mobile : +94 77 804 8154
Email : asintha.99@outlook.com
LinkedIn : https://www.linkedin.com/in/asintha-ariyarathne-54450b245/
Website : asi-a99.github.io

PERSONAL PROFILE

I am a motivated engineering graduate with a solid foundation in electrical and electronic engineering and computer science. I enjoy solving complex problems with practical, innovative solutions. Hands-on team projects have strengthened my communication, teamwork, and project management skills. I'm eager to keep learning and contribute to a forward-thinking organization.

EDUCATION

Bachelor of the Science of Engineering (BSc.Eng) Graduate of University of Peradeniya
Specialization: Electrical and Electronic Engineering - [2021 to 2025]
Second Class Honours

Secondary Education:

Kingswood College, Kandy: Grade 6 - 13 [2010 - 2018]

- G.C.E. (A/L) Examination in Physical Science Stream - 2019: 2 A's 1 B (Combined mathematics) | District's rank 10th
- G.C.E. (O/L) Examination - 2016 : 7 A's 1 C(Tamil) 1 S(English literature)

Primary Education:

Sri Rahula College, Katugasthota : Grade 1 - 5 [2006 - 2010]

- Grade 5 Scholarship Examination - 2010 : 176 marks | District's 32nd

PROFESSIONAL EXPERIENCE

Embedded Software Engineer [September 2026 - October 2026]

- Thakshana Technologies (Pvt) Ltd.
  - Built custom Yocto Project Linux images, custom meta-layers, and cross-compilation SDK toolchains for 64-bit ARM networking processors (NXP LS1088A).
  - Customized early-boot firmware stacks including RCW/PBI, TF-A, and U-Boot and performed schematic-level hardware validation involving CPLD boot-muxing logic and QSPI/SD flash interfaces.
  - Developed and integrated Linux kernel device drivers and device tree configurations, implemented secure boot across the boot chain.

Electronic Engineer [October 2025 - September 2026]

- Vega Innovations (Pvt) Ltd.
  - Developed high-efficiency control firmware using interrupt architectures and finite state machines (FSMs) to enhance real-time performance, system stability, and reliability.
  - Implemented SPI, SCI, and CAN communication protocols for reliable data exchange between embedded systems and peripheral devices.
  - Designed multi-layer PCBs with a strong focus on signal integrity, reducing electromagnetic interference (EMI), eliminating ground loops, and improving thermal dissipation for high-power applications.

- Performed comprehensive hardware testing and advanced troubleshooting using precision oscilloscopes, logic analyzers, and power analyzers to verify performance and ensure full design compliance.Performed thermal simulation on the PCB design to validate the heatsink and forced-air cooling strategy under worst-case operating conditions.

Electrical Engineering Intern [August 2024 - October 2024]

- South Asia Gateway Terminals (SAGT)
  - Performed troubleshooting and repairs of electrical systems in quay cranes and rubber-tyred gantry cranes. (Cable reel/motor repairing, limit sensor/ proximity sensor/ encoder troubleshooting and etc.)
  - Developed Ladder, STL, and FBD programs for PLCs. (To bypass the safety protocols of Quay cranes under emergency situations to have gantry movements.)
  - Contributed to multiple industrial automation and energy optimization projects, including upgrading a pump house with PLCs for enhanced control and conducting a feasibility study on installing a solar power system on RTG girders to improve energy efficiency.

Electrical Engineering Intern [August 2023 - October 2023]

- Diesel and Motor Engineering PLC
  - Successfully coordinated the 21.09 kW solar photovoltaic installation project in its entirety.
  - Engaged in numerous generator service repairs and troubleshooting electrical machinery. (Winding Reinsulation , Megger testing, Generator controller Configuration and etc.)
  - Created multiple industrial and procurement reports using MS Word and MS Excel.

Physics Tutor (Part-Time) [Since 2021 - present]

- AA_Physics (Self owned)
  - Followed well-planned and organized teaching sessions for local and Edexcel syllabuses.
  - Helped students stay motivated and develop a strong interest in learning.
  - Gained valuable experience in planning, leadership, and clear communication through teaching.

PROJECTS

Edge Router | Embedded Linux Networking Platform (Ongoing)

Designed and built a working edge router on a BeagleBone Black from the bootloader up, booting a self-built U-Boot and Linux kernel over TFTP. Configured routing and NAT behind an nftables firewall with VLAN-separated networks, and dynamic routing with FRR (OSPF and BGP). Developing a netfilter kernel module, eBPF/XDP packet filters, an rtnetlink link-monitoring tool in C cross-compiled for ARM. Building a pytest suite that tests the real firewall in Linux network namespaces and on the hardware, with GitHub Actions CI, a lintian-checked Debian package and DKMS packaging for the kernel module.

Skills gained – Embedded Linux, Linux Kernel Modules, U-Boot, eBPF/XDP, Netfilter and nftables, FRR (OSPF, BGP), Netlink, C, Python (pytest), Debian Packaging, DKMS.

Off-Grid Solar Inverter | Low-Cost High-Efficiency Power Conversion System

Designed and developed a low-cost off-grid solar inverter consisting of an interleaved push-pull DC-DC converter and a full-bridge inverter stage for efficient power conversion. Performed magnetic component design, including transformer and inductor calculations and optimization for high-efficiency operation. Selected suitable power and control ICs, developed the complete schematic for reliable hardware implementation. Validated converter operation and control behavior through simulation using PLECS and TINA-TI.

Skills gained – Magnetic Component Design, Power Electronics Design, Schematic Design, PCB Design, PLECS, TINA-TI, DC-DC Converters, Full-Bridge Inverters, and Hardware Development.

Three-Level NPC Inverter Using GaN Devices | High-Efficiency Solar PV Inverter

Designed and developed a high-efficiency single-phase Three-Level NPC inverter using GaN power devices for solar PV applications. Implemented a cascaded dual-loop PI controller, mathematically derived and validated through PLECS simulation. Developed the complete firmware stack on a TI C2000 MCU, incorporating CLA-based real-time control, a state machine, fault detection, and SPI communication. Resolved hardware issues including GaN gate-drive noise, Miller-effect coupling, and PCB ground-bounce through systematic diagnosis and a full PCB redesign. Performed thermal simulation to validate the cooling strategy under worst-case operating conditions.

Skills gained - PLECS, TI C2000, C Programming, PCB Layout (Fusion 360), Control Systems, Embedded Firmware Development, Power Electronics Hardware Troubleshooting, and Thermal Simulation.

Microinverter for CdTe/CdS Solar Module | Clean Accessible Energy for All Sri Lankans

Designed an off-grid micro inverter tailored for Sri Lankan-manufactured CdTe/CdS solar modules. Integrated a load tracking algorithm and an energy storage system (battery) with a battery management system (BMS). The final product combined the micro inverter and energy storage device into a single compact unit within the solar module, optimized for off-grid applications.

Skills gained - MATLAB/Simulink, PSIM, PLECS, Altium Designer, thermal design, battery management systems, control systems, and electronic circuit troubleshooting

LICENSES & CERTIFICATIONS

- Introduction to Linux (LFS101) - The Linux Foundation
- Artificial Intelligence Fundamentals - IBM
- Introduction to Data Science - Cisco
- Using Python to Access Web Data - Coursera
- Blockchain: Foundations and Use Cases - Coursera
- Engineering drawing and 3D modelling using AutoCAD - tecviewonline
- Programming for Everybody (Getting Started with Python) - Coursera
- Python Data Structures - Coursera

VOLUNTEER WORK

- Institution of Engineers of Sri Lanka (IESL) - Associate Member
- Electrical and Electronic Engineering Society (EEES) - Member
- Gavel club of University of Peradeniya - Committee Member
- Rotaract Club of University of Peradeniya - Member
  - Contributed as a teacher to the Ordinary Level Seminar Series held at Rasindev Vidyalaya, Kandy.
- Kingswood Astronomical Society - Committee Member
- Kingswood Science Society - Member

REFERENCES

Prof. Lilantha Samaranayake
Senior Professor,
Department of Electrical & Electronic Engineering,
Faculty of Engineering,
University of Peradeniya,
Sri Lanka.
lilantha@eng.pdn.ac.lk
+94 71 684 8319

Mr. Lasitha Perumpuliarachchi
Manager,
Electrical & Electronic Engineering,
Department of Engineering Solutions,
Diesel & Motor Engineering PLC,
Colombo,
Sri Lanka.
lasitha.yasas@dimolanka.com
+94 77 355 1134